SPECTROPHOTOMETRY

When a beam of monochromatic light of radiation is passed through a solution, its intensity goes on decreases. This is because of the light corresponding to a certain frequency is absorbed by the solution. If I_0 is the intensity of the incident beam of light and I_t is the intensity of the transmitted beam, then the intensity of light absorbed I_a is given by $I_a = I_0 - I_t$

Lambert- Beer Law and derivation of its mathematical form:

When a monochromatic beam of light passes through a solution of an absorbing substance , the rate of decrease of intensity of radiation with the thickness of the absorbing solution is proportional to the intensity of radiation and the concentration of the solution .

Mathematically , $-\frac{\partial I}{\partial x} \propto c\, I$ OR $\frac{\partial I}{I} = -kc\, \partial x$ .....(1)

Where '∂I' is the change in the intensity produced by the absorption of radiation on passing a thickness '∂x' of solution of concentration 'c' and 'k' is the proportionality constant . The minus sign indicates the reduction in intensity.

Integrating the equation (1), between the limits $I=I_0$ at x=0 and I = I at x=x, we get

$$\int_{I_0}^{I} \frac{\partial I}{I} = -kc \int_{0}^{x} \partial x$$

i.e $$\ln \frac{I}{I_0} = -kcx \quad \text{..... (2)}$$

Or $$\frac{I}{I_0} = e^{-kcx} \quad \text{..... (3)}$$

Eqn (3) is called the Mathematical form of Lambert – Beer law.

Definition: *The intensity of a beam of monochromatic radiation decreases exponentially with increase in the thickness 'x' and the concentration 'c' of the absorbing medium. This is called Lambert –Beer Law .*

In its mathematic form $\boldsymbol{\frac{I}{I_0} = e^{-kcx}}$

Molar / Molal extinction coefficient :

Consider the equation(2), $\ln\frac{I}{I_0} = -kcx$ or $2.303\log\frac{I}{I_0} = -kcx$

$$\log\frac{I}{I_0} = \frac{-kcx}{2.303}$$

Substituting $\frac{k}{2.303} = \alpha$, we got $\log\frac{I}{I_0} = -\alpha cx$ .....

....(4). Where 'α' is called Molar Extinction coefficient or absorbivity.

The eqn (4) can also be written as, $\log\frac{I_0}{I} = \alpha cx$ *OR* $\alpha = \frac{1}{cx}\log\frac{I_0}{I}$

If concentration c=1M and $\log\frac{I_0}{I}$ =1, then $\alpha = \frac{1}{x}$

$\log\frac{I_0}{I} = 1$ *means* $\frac{I_0}{I} = 10$

Or $I_0 = I \times 10$

Or $I = \frac{1}{10} \times I_0$ .....(A)

Definition of Molar / molal extinction coefficient or Molal Absorbitivity:

The molar / molal extinction coefficient may be defined as the reciprocal of that thickness of the solution layer of one molar / molal concentration which reduces the intensity of the light passing through it to one tenth of its original value.

The molar extinction coefficient is the characteristic of the solute and depends on the solvent, temperature and wavelength of the radiation used.

Application of Lambert's –Beer Law:

i. In spectrometer's to find the molal extinction coefficient.
ii. In colorimeters to find the optical density and absorption.
iii. In gravimetric and volumetric analysis.

Limitations of Lambert's –Beer Law:

I. The law is applicable to dilute solutions only.
II. The radiation used to be only monochromatic.
III. The temperature of the system during the experiment to be maintained constant.

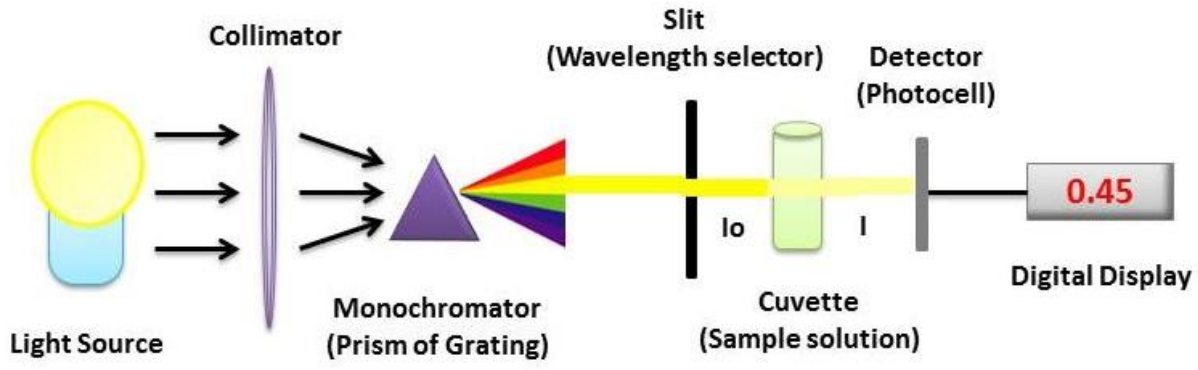
Problems:

1. A monochromatic beam of radiation is passed through a 0.05M solution. The intensity of the radiation is reduced to one fourth of the initial value after passing through 10 cm length of the solution. Calculation the molar extinction coefficient of the substance.
   Solution: given c=0.05M , x= 10cm. (let I_0=100 then I will be 25)
   $\alpha = \frac{1}{cx}\log\frac{I_0}{I}$
   $\alpha = \frac{1}{0.05\times10}\log\frac{100}{25} \Rightarrow \alpha = \frac{1}{0.5}\log 4 = 0.06021\times2 = 1.2042\ mol^{-1}cm^{-1}$

2. A substance when dissolved in water at 10^{-3} M concentration absorbs 10% of an incident radiation in a path of 1cm length . what should be the concentration of the solution in order to absorb 90% of the same radiation?
   Solution: $\alpha = \frac{1}{cx}\log\frac{I_0}{I}$
   $\alpha = \frac{1}{10^{-3}\times1}\log\frac{100}{90} \Rightarrow \alpha = 45.75\ mol^{-1}cm^{-1}$

   $\alpha = \frac{1}{cx}\log\frac{I_0}{I}$
   $45.75 = \frac{1}{cx}\log\frac{100}{10} \Rightarrow \alpha = \frac{1}{45.75}\log 10 = \frac{1}{45.75}\times1 = 0.0218\ mol$

   Spectrophotometer : The absorption spectra of solution can be studied with the help of an instrument called spectrophotometer.
   The instrument used is as shown in the figure.



   The radiation from a source of mercury lamp is passed through a condensing lens and then through a narrow slit to select a narrow beam. This beam is made to fall on a mirror. The reflected beam is made to fall on a

grating. By rotating the grating , light of different wavelengths can be selected.

The light from the prism is made to pass through a suitable cell. The emerging light from the cell is made to fall on the cathode of a photoelectric cell. The photo electricity generated is measured by using a galvanometer. In actual practice , light is passed through the cell containing pure solvent and the intensity of the emerged beam(I_0) is noted. The solvent in the cell is replaced with the solution and the intensity of the emerged beam (I) is noted. The photo electricity generated is proportional to the intensity. The experiment is repeated for different wavelengths .

By knowing the thickness 'x' of the solution through which light has passed and the concentration 'c' the molal extinction coefficient 'α' can be calculated using the equation $\log\frac{I_0}{I} = \alpha\, cx$.

The molal extinction coefficient values are plotted against wavelength. The curve obtained is called absorption band.

Application of Molal extinction coefficient :

1. It helps to identify the given substance , since every substance will have its characteristic value.
2. Purity of the substance can be tested as pure substance has definite value which is altered by the impurities.
3. The presence of functional groups in an organic compound can be predicted as the specific groups in organic compounds are associated with specific absorption bands.